

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY**Bachelor of Science in Information Technology (B.Sc.IT)****Semester-II****University Examination May-2018****CA 409 – Computer Oriented Numerical Methods****Date: 11/05/2018, Friday****Time: 10:00 a.m. to 01:00 p.m.****Maximum Marks: 70****Instructions:**

1. Attempt the two sections in separate answer books.
2. Figure to the right indicate marks.
3. Make suitable assumptions wherever necessary and state them.

SECTION-I**Q.1 Do as directed :**

1. Newton Backward interpolation formula is used for _____ intervals. [1]
A. equal B. unequal C. open D. closed
2. Differentiate between Gauss-Elimination and Gauss-Jordan Method. [2]
3. Construct the difference table for $f(x)=(x+2)^2$ for $x=3,4,5,6$ and find $\Delta^2 f(4)$. [2]
4. Solve $x^2 - 2 = 0$ by Newton-Raphson technique. If initial guess is $X_0 = 1.0$, subsequent estimate of x (i.e. x_1) = _____. [2]
5. The _____ method is limited to LP problems involving two decision variables and a limited number of constraints due to the difficulty of graphing and evaluating more than two decision variables. [1]
6. Match the following: [3]

A. Newton-Raphson	1. Integration
B. Runge-kutta	2. Root finding
C. Simpson's Rule	3. Ordinary Differential Equations

Q.2[A] Evaluate $f(x) = x^2 - 2x - 2$ using Bisection Method up to four steps. [5]**[B] From the following table of yearly premiums for policies maturing at quinquennial ages, estimate the premiums for policies maturing at the age of 46 years using Newton Forward Method. [7]**

Age(x)	45	50	55	60	65
Premium(y)	2.871	2.404	2.083	1.862	1.712

OR**Q.2[A] Evaluate $f(x) = x^3 + x^2 - 3x - 3$ using Regula Falsi Method up to four steps. [5]****[B] The population of a town in the last six censuses was as given below. Estimate the population for the year 1946 using Newton Backward Method. [7]**

Year(x)	1911	1921	1931	1941	1951	1961
Population(thousands)	12	15	20	27	39	52

Q.3[A] Evaluate $\int_0^1 \frac{dx}{(2x-1)}$ by Simpson's one-third rule, taking four sub-intervals. **[5]**

[B] Find the value of $\nabla^2 f(5)$ for the following table of values for $f(x)$ given below. **[3]**

x	0	5	10	15
f(x)	1.0	1.6	3.8	6.3

[C] Compute $f(0.4)$ for the table below by Lagrange's interpolation. **[4]**

x	0.3	0.5	0.6
f(x)	0.61	0.69	0.72

OR

Q.3[A] Evaluate $\int_0^1 \frac{dx}{1+x^2}$ using Trapezoidal Rule. **[6]**

[B] Determine the root of the equation, $x^2 - 6x + 2$, by Newton-Raphson method taking $X_0 = 6$. **[6]**

SECTION-II

Q.4[A] Do as directed: **[8]**

- Simpson's 3/8th rule is applicable only when _____.
A. n is a multiple of 3 B. n is a multiple of 6
C. n is a multiple of 8 D. n is a multiple of 2
- _____ is used to denote the process of finding the values not inside the interval (X_0, X_n) .
A. Interpolation B. Extrapolation
C. Iterative D. Polynomial equation
- To find the derivative for the end value (lies between) of the table _____ formula is used.
A. Newton Forward Interpolation Formula
B. Newton Backward Interpolation Formula
C. Newton Forward Difference Formula
D. Newton Backward Difference Formula
- The Forward difference interpolation operation is denoted by the symbol _____.
- Which are correct initial values of $f(x) = x^2 - 3x - 3$ to find the root of nonlinear equation?
- In Regula-Falsi Method, which formula is used to find out first approximation value?
- _____ interpolation method is suitable to find $f(5.6)$ if $x = 1, 3, 4, 6, 9$
- The degree of $y(x)$ in Trapezoidal Rule is _____.
A. 1 B. 2 C. 3 D. 6

[B] Differentiate between REF and RREF Method. **[3]**

Q.5[A] Using Range-Kutta Method of forth order solve: **[6]**

$\frac{dy}{dx} = 3x + \frac{1}{2}y$ with $y(0) = 1$ at $x = 0.2$, taking $h = 0.1$

- [B] Use Euler's Method to approximate y when $x=0.4$, given that $\frac{dy}{dx}=x+2y, y(0)=0, h=0.1$ [6]

OR

- Q.5[A] How to formulate LPP problem? Explain with example. [7]

- [B] Using Gauss-Jordan method, solve the following system of equations [5]

$$x + y + z = 4, 2x - 3y + z = 2, -x + 2y - z = -1$$

- Q.6 Answer the following.(Any three) [12]

- [A] Solve the following LPP using Graphical Method.

$$\text{Maximize } Z=5x+10y$$

Subject to Constraints:

$$5x+8y \leq 40$$

$$3x+y \leq 12$$

$$x, y \geq 0$$

- [B] Solve the system of equations $x+y+z=3; 2x+3y+7z=0; x+3y-2z=17$ using Gauss – elimination method.
- [C] Using Taylor series method, find $y(0.1)$ for $y'=x-y^2, y(0)=1$ correct up to three decimal places.
- [D] Use Euler's Method to approximate y when $x=0.3$, given that $\frac{dy}{dx}=x+y, y(0)=1, h=0.1$
- [E] Convert following matrix into RREF.

A.

$$\begin{bmatrix} 1 & 2 & 1 \\ -1 & 0 & 2 \\ 2 & 3 & -3 \end{bmatrix}$$

B.

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{bmatrix}$$